Correlated HOPS for Propagating Structured Environments

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I. INTRODUCTION

The hierarchy of pure states (HOPS) is a stochastic wavefunction method for open quantum systems coupled linearly to Gaussian environments with a finite memory time [1, 2, 3, 4]. It avoids a Born-Markov approximation and, unlike a direct system-plus-environment calculation, does not require a Fock-space truncation for every retained bosonic mode. In our earlier work we combined nonlinear HOPS with matrix product states (MPS) to simulate non-Markovian dynamics in strongly interacting one-dimensional systems [6]. That construction attached an independent damped environmental mode to each system site. The paper is available at Phys. Rev. Lett. 128, 063601 (2022).

Independent local environments are a useful starting point, but they exclude a simple and widespread piece of physics: an excitation emitted by one site can propagate through a structured environment and later influence another site. This occurs in coupled cavity arrays, multimode and waveguide cavity QED, circuit resonators, phononic and magnonic networks, and impurities embedded in a common reservoir. After those environmental degrees of freedom are eliminated, the system sees a matrix-valued memory kernel $\alpha_{ij}(t-s)$ rather than a set of diagonal correlations. The local noises are correlated in both space and time, and the hierarchy response must carry the same correlations.

This note derives a correlated version of HOPS for a quadratic bosonic environment. A matrix-exponential kernel can be represented either by independent collective normal-mode channels or by local hierarchy directions whose auxiliary occupations propagate between sites. These are two bases for the same memory. We also derive the corresponding nearly unitary HOPS (nuHOPS) formulation [9]. A trajectory-dependent displacement keeps the auxiliary memory close to its vacuum while preserving the complete matrix-valued bath drift. We then use normalized nonlinear HOPS, including its Girsanov noise shift, to remove two coherently coupled damped photon modes from a two-qubit problem. The modes are retained only in a full master equation used as an independent benchmark.

The numerical example is deliberately transparent. Qubit 1 emits into its local photon mode, the excitation travels to the second mode, and qubit 2 absorbs it. At later times the flow reverses. We treat both the excitation-preserving Jaynes-Cummings interaction and the full quantum-Rabi interaction. In addition to reduced qubit observables, HOPS reconstructs the photon occupations, hopping current, inter-mode correlations and every qubit-mode two-point correlation without retaining a photon Hilbert space. The same construction can be inserted into the MPS framework of Ref. [6], providing a route to interacting many-body systems coupled through propagating structured environments.

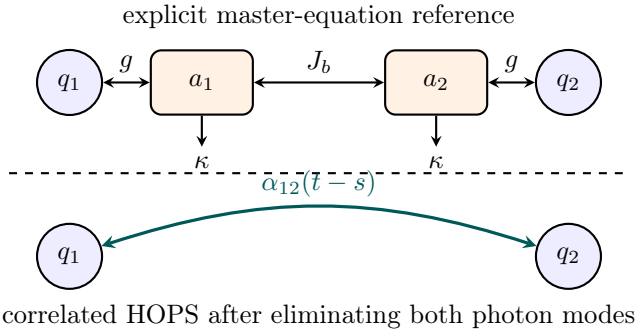


FIG. 1. The benchmark and its reduced description. In the enlarged model, each qubit couples only to its local mode and the modes hop coherently. After the modes are traced out, the qubits are driven by a matrix-valued coloured noise and memory kernel. No direct qubit-qubit coupling is introduced.

II. QUADRATIC ENVIRONMENTS AND THE CORRELATION MATRIX

Consider N system coupling operators L_i and M bosonic coordinates b_μ. The total Hamiltonian is

$$H_{\rm tot} = H_S + \sum_{\mu\nu} b_\mu^\dagger h_{\mu\nu} b_\nu + \sum_{i\mu} \left(L_i V_{i\mu} b_\mu^\dagger + L_i^\dagger V_{i\mu}^* b_\mu \right), \tag{1}$$

where $h = h^\dagger$. We assume an initially factorized state, a vacuum Gaussian environment and linear system-bath coupling. Markovian loss of the bosonic coordinates may be included through a positive photon-number damping matrix K. The free Heisenberg amplitudes then propagate with

$$W = \frac{K}{2} + ih, \qquad \boldsymbol{b}(t) = e^{-Wt}\boldsymbol{b}(0) + \text{input noise}. \tag{2}$$

The factor $K/2$ is important: if a photon population decays at rate κ, its field amplitude and vacuum correlation decay at rate $\kappa/2$.

Define the bath fields $B_i = \sum_\mu V_{i\mu} b_\mu$. For $\tau \geq 0$ their vacuum correlation matrix is

$$\boxed{\alpha(\tau) = \mathbb{E}[\boldsymbol{B}(\tau)\boldsymbol{B}^\dagger(0)] = Ve^{-W\tau}V^\dagger.} \tag{3}$$

The continuation to negative time is $\alpha(-\tau) = \alpha(\tau)^\dagger$. Equation (3) contains the onsite spectral density, the propagation between bath coordinates and the spatial profile of the system-bath coupling. In particular, $\alpha_{ij}(0) = 0$ does not imply that $\alpha_{ij}(\tau)$ vanishes at later times.

For the two local photon modes in Fig. 1,

$$h_B = \begin{pmatrix} \omega_c & J_b \\ J_b & \omega_c \end{pmatrix}, \qquad K = \kappa I, \qquad V = gI. \tag{4}$$

The symmetric and antisymmetric modes have frequencies $\omega_\pm = \omega_c \pm J_b$. Direct exponentiation gives

$$\boxed{\alpha(\tau) = g^2 e^{-(\kappa/2 + i\omega_c)\tau} \begin{pmatrix} \cos(J_b\tau) & -i\sin(J_b\tau) \\ -i\sin(J_b\tau) & \cos(J_b\tau) \end{pmatrix}.} \tag{5}$$

The local modes are independent at equal times but become correlated as the field propagates. This is qualitatively different from imposing a fixed spatial covariance matrix multiplying one scalar temporal function.

III. NON-MARKOVIAN QUANTUM STATE DIFFUSION

Expand the total state in unnormalized Bargmann coherent states of the bath,

$$|\Psi_{\text{tot}}(t)\rangle = \int \frac{\mathrm{d}^{2M}z}{\pi^M} e^{-|z|^2} |\psi_t(z^*)\rangle \otimes |z\rangle. \tag{6}$$

The coherent-state relations replace a creation operator by multiplication with z_μ^* and an annihilation operator by a derivative with respect to z_μ^*. Applying the chain rule to the effective coloured noises produces the multi-channel NMQSD equation

$$\begin{aligned} \partial_t|\psi_t\rangle = & \left[-iH_S + \sum_i L_i z_i^*(t)\right]|\psi_t\rangle \\ & - \sum_{ij} L_i^\dagger \int_0^t \mathrm{d}s\, \alpha_{ij}(t-s) \frac{\delta|\psi_t\rangle}{\delta z_j^*(s)}. \end{aligned} \tag{7}$$

The proper complex Gaussian noises obey

$$\mathbb{E}[z_i(t)z_j^*(s)] = \alpha_{ij}(t-s), \qquad \mathbb{E}[z_i(t)z_j(s)] = 0. \tag{8}$$

The reduced density operator is $\rho_S(t) = \mathbb{E}[|\psi_t\rangle\langle\psi_t|]$. Equation (7) is nonlocal in time because of its functional derivative. HOPS replaces this functional memory by a time-local hierarchy of auxiliary pure states.

IV. THE CORRELATED HIERARCHY OF PURE STATES

A. Separable exponential kernels

First suppose

$$\alpha_{ij}(\tau) = C_{ij}e^{-\nu\tau}, \qquad \tau \geq 0, \tag{9}$$

where $C = C^\dagger \succeq 0$. Define

$$D_i(t) = \sum_j \int_0^t \mathrm{d}s\, C_{ij} e^{-\nu(t-s)} \frac{\delta}{\delta z_j^*(s)}, \qquad |\psi^{\boldsymbol{k}}\rangle = \prod_i D_i^{k_i}|\psi\rangle. \tag{10}$$

Using $\dot{D}_i = -\nu D_i$ and $[D_i, z_j^*(t)] = C_{ij}$ gives

$$\partial_t|\psi^{\boldsymbol{k}}\rangle = \left[-iH_S - |\boldsymbol{k}|\nu + \sum_j L_j z_j^*(t)\right]|\psi^{\boldsymbol{k}}\rangle + \sum_i k_i \left(\sum_j C_{ij}L_j\right)|\psi^{\boldsymbol{k}-\boldsymbol{e}_i}\rangle - \sum_i L_i^\dagger|\psi^{\boldsymbol{k}+\boldsymbol{e}_i}\rangle. \tag{11}$$

Spatial covariance therefore enters twice: in the coloured noise and in the lowering response of the hierarchy. Correlating the noises while leaving a local term $k_i C_{ii} L_i$ does not solve the same open-system problem.

If $C = RR^\dagger$, one can instead introduce independent collective channels $A_r = \sum_i R_{ir}^* L_i$. Standard independent-channel HOPS for the A_r is exactly equivalent to Eq. (11). This transformation is particularly efficient when C has low rank [5].

B. Propagating local memory

The cavity kernel is a matrix exponential rather than Eq. (9). For

$$\alpha(\tau) = e^{-W\tau}C, \tag{12}$$

the derivative vector satisfies $\dot{D}_i = -\sum_j W_{ij}D_j$. The hierarchy becomes

$$\partial_t|\psi^{\boldsymbol{k}}\rangle = \left(-iH_S + \sum_i L_i z_i^*\right)|\psi^{\boldsymbol{k}}\rangle - \sum_{ij} k_i W_{ij}|\psi^{\boldsymbol{k}-\boldsymbol{e}_i+\boldsymbol{e}_j}\rangle + \sum_i k_i \left(\sum_m C_{im}L_m\right)|\psi^{\boldsymbol{k}-\boldsymbol{e}_i}\rangle - \sum_i L_i^\dagger|\psi^{\boldsymbol{k}+\boldsymbol{e}_i}\rangle. \tag{13}$$

The off-diagonal elements of W move an auxiliary occupation between hierarchy directions while preserving its total tier. For a local bath hopping matrix this has a local tensor-network form. After the rescaling $\chi^{\boldsymbol{k}} = \psi^{\boldsymbol{k}}/\sqrt{\prod_i k_i!}$, introduce auxiliary bosons c_i. For $C = I$ the generator contains

$$\mathcal{G}(t) = -iH_S + \sum_i L_i z_i^* + \sum_i (L_i c_i^\dagger - L_i^\dagger c_i) - \sum_{ij} W_{ij} c_i^\dagger c_j. \tag{14}$$

Grouping each system site with its local hierarchy coordinate therefore extends the MPS layout of Ref. [6]: the physical environment has been traced out, while its memory propagates through the auxiliary chain. The hierarchy indices are computational memory coordinates, not physical photon Fock states.

C. Collective normal modes and nonlinear HOPS

For the present two-mode calculation, K is proportional to the identity and h_B is diagonalized by a unitary U. We use independent normal-mode hierarchy directions with

$$S_r = \sum_i U_{ir}^* S_i, \qquad \nu_r = \frac{\kappa}{2} + i\omega_r, \qquad r \in \{+,-\}, \tag{15}$$

where $S_i = \sigma_i^-$ for Jaynes-Cummings coupling and $S_i = \sigma_i^x$ for Rabi coupling. The channel strength g is kept explicit.

Sampling variance is reduced with normalized nonlinear HOPS. Let $s_r(t)$ be the Girsanov history shift,

$$\dot{s}_r = -\nu_r^* s_r + g\langle S_r^\dagger\rangle_t, \qquad s_r(0) = 0, \tag{16}$$

and let $z_r(t)$ be independent stationary complex Ornstein-Uhlenbeck processes with unit variance and correlation $\mathbb{E}[z_r(t)z_s^*(u)] = \delta_{rs}e^{-\nu_r(t-u)}$ for $t \geq u$. In a normalized root-state gauge the hierarchy has the form

$$\partial_t|\phi^{\boldsymbol{k}}\rangle = \left[-iH_S - \sum_r k_r\nu_r + g\sum_r S_r(z_r^* + s_r) + \chi_t I\right]|\phi^{\boldsymbol{k}}\rangle + g\sum_r k_r S_r|\phi^{\boldsymbol{k}-\boldsymbol{e}_r}\rangle - g\sum_r \left(S_r^\dagger - \langle S_r^\dagger\rangle_t\right)|\phi^{\boldsymbol{k}+\boldsymbol{e}_r}\rangle. \tag{17}$$

The scalar gauge χ_t keeps the root normalized and does not change physical ensemble averages. No single-excitation closure is used in the calculations below: Eq. (17) is propagated unchanged for both interactions.

V. NEARLY UNITARY HOPS FOR CORRELATED ENVIRONMENTS

Normalized nonlinear HOPS improves the sampling measure, but a highly excited environment can still require many hierarchy tiers. nuHOPS addresses this separate problem by following the coherent displacement of the auxiliary memory and expanding only its residual fluctuations around a moving origin [9]. The displacement is a change of hierarchy basis: it does not alter the physical root trajectory or the bath kernel.

A. Shifted auxiliary hierarchy

It is useful first to write the matrix kernel in a whitened coordinate form,

$$\alpha(\tau) = Re^{-W\tau}R^\dagger, \qquad \tau \geq 0. \tag{18}$$

Introduce coordinate noises $\boldsymbol{x}$ with $\mathbb{E}[\boldsymbol{x}(t)\boldsymbol{x}^\dagger(s)] = e^{-W(t-s)}$ for $t \geq s$, and define the system operators

$$A_q = \sum_i R^*_{iq} L_i. \tag{19}$$

Then $\sum_q A_q x^*_q$ is exactly the site-noise term with covariance Eq. (18). This factorization includes the cavity model with $R = gI$ and also a low-rank or fitted correlated kernel.

Let $\boldsymbol{D}$ denote the corresponding HOPS memory derivatives in these coordinates, so that

$$\dot{\boldsymbol{D}} = -W\boldsymbol{D}. \tag{20}$$

The normalized nonlinear root obeys, up to a scalar normalization gauge,

$$\partial_t|\phi^{\mathbf{0}}\rangle = \left[-iH_S + \sum_q \widetilde{x}^*_q A_q\right]|\phi^{\mathbf{0}}\rangle - \sum_q \left(A^\dagger_q - \langle A^\dagger_q\rangle_0\right) D_q|\phi^{\mathbf{0}}\rangle + \chi_t|\phi^{\mathbf{0}}\rangle, \tag{21}$$

where the tilde denotes the Girsanov-transformed noise and every expectation value is evaluated in the physical root.

For an initially vacuum bath, choose a time-dependent displacement $\boldsymbol{\mu}(0) = 0$ and define factorial-scaled, displaced auxiliary states

$$|\phi^{\boldsymbol{k}}_\nu\rangle = \prod_q \frac{(-i)^{k_q}}{\sqrt{k_q!}}\left(D_q - \mu_q\right)^{k_q}|\phi^{\mathbf{0}}\rangle, \qquad |\Phi_\nu\rangle = \sum_{\boldsymbol{k}} |\phi^{\boldsymbol{k}}_\nu\rangle \otimes |\boldsymbol{k}\rangle. \tag{22}$$

The kets $|\boldsymbol{k}\rangle$ are Fock states of computational auxiliary bosons b_q; they are not photon states. Differentiating $\boldsymbol{D} - \boldsymbol{\mu}$ gives

$$\partial_t(\boldsymbol{D} - \boldsymbol{\mu}) = -W(\boldsymbol{D} - \boldsymbol{\mu}) - (\dot{\boldsymbol{\mu}} + W\boldsymbol{\mu}). \tag{23}$$

With $\boldsymbol{m} = \dot{\boldsymbol{\mu}} + W\boldsymbol{\mu}$, the displaced hierarchy is therefore

$$\begin{aligned}\partial_t|\Phi_\nu\rangle = &\left[-iH_S - \sum_{qp} W_{qp} b^\dagger_q b_p + \sum_q \left(\widetilde{x}^*_q A_q - \mu_q A^\dagger_q\right) + \chi_t I\right]|\Phi_\nu\rangle \\ &- i\sum_q \left[(A_q - m_q)b^\dagger_q + \left(A^\dagger_q - \langle A^\dagger_q\rangle_0\right) b_q\right]|\Phi_\nu\rangle.\end{aligned} \tag{24}$$

Equation (24) is exact for any differentiable $\boldsymbol{\mu}(t)$ when the hierarchy is untruncated. The freedom in $\boldsymbol{\mu}$ is used to reduce the truncated auxiliary occupation.

B. Mean-field displacement and near-unitary form

The mean-field choice

$$\boxed{\dot{\boldsymbol{\mu}} = -W\boldsymbol{\mu} + \langle \boldsymbol{A}\rangle_0, \qquad \boldsymbol{\mu}(0) = 0} \tag{25}$$

makes $\boldsymbol{m} = \langle \boldsymbol{A}\rangle_0$. It is the multi-coordinate extension of the nuHOPS shift of Müller and Strunz and makes $\langle b_q\rangle \simeq 0$ a stable mean-field fixed point. The same convolution appears in the nonlinear sampling transformation,

$$\widetilde{x}^*_q = x^*_q + \mu^*_q. \tag{26}$$

Thus the Girsanov shift used for variance reduction also supplies the moving auxiliary origin; nuHOPS does not replace nonlinear importance sampling.

Write

$$W = \frac{K}{2} + ih, \qquad K = W + W^\dagger \succeq 0, \qquad h = h^\dagger. \tag{27}$$

Substituting Eqs. (25) and (26) into Eq. (24) gives the correlated nuHOPS generator

$$\begin{aligned}\partial_t|\Phi_\nu\rangle = &-i\Bigg[H_S+\sum_{qp}h_{qp}b_q^\dagger b_p+i\sum_q\left(\mu_q^*A_q-\mu_qA_q^\dagger\right)\\ &\qquad+\sum_q\left\{\left(A_q-\langle A_q\rangle_0\right)b_q^\dagger+\left(A_q^\dagger-\langle A_q^\dagger\rangle_0\right)b_q\right\}\Bigg]|\Phi_\nu\rangle\\ &+\left[\sum_q x_q^*A_q-\frac{1}{2}\sum_{qp}K_{qp}b_q^\dagger b_p+\chi_tI\right]|\Phi_\nu\rangle. \end{aligned}\tag{28}$$

The first two lines are generated by a Hermitian operator. Only the coloured noise, positive auxiliary damping and the scalar normalization gauge are not of Hamiltonian form, hence the name nearly unitary HOPS. The physical trajectory is the auxiliary vacuum projection $|\phi^{\mathbf{0}}\rangle=\langle\mathbf{0}|\Phi_\nu\rangle$, and the reduced state is the same ensemble average of normalized roots as in nonlinear HOPS.

C. Collective modes versus propagating local memory

For the two coupled cavities, diagonalizing W gives the two independent collective channels already used in Eq. (15),

$$\alpha_{ij}(\tau)=g^2\sum_{r=\pm}U_{ir}e^{-\nu_r\tau}U_{jr}^*,\qquad A_r=gS_r.\tag{29}$$

Each channel has its own displaced auxiliary oscillator and

$$\dot\mu_r=-\nu_r\mu_r+g\langle S_r\rangle_0,\qquad \mu_r=s_r^*.\tag{30}$$

This is the direct multi-channel form of published nuHOPS.

Alternatively, retain local bath coordinates. Then $A_i=gS_i$ and $W=\kappa I/2+ih_B$, so that

$$\dot\mu_i=-\frac{\kappa}{2}\mu_i-i\sum_j(h_B)_{ij}\mu_j+g\langle S_i\rangle_0.\tag{31}$$

The term $\sum_{ij}(h_B)_{ij}b_i^\dagger b_j$ in Eq. (28) transports the displaced hierarchy memory between local auxiliary sites, while $\kappa\sum_i b_i^\dagger b_i/2$ damps it. Equations (30) and (31) are related by the same unitary transformation as the physical photon modes. The collective basis minimizes channel coupling; the local basis preserves spatial locality for an MPS implementation.

The figures below were generated with normalized nonlinear HOPS, not nuHOPS. Since nuHOPS is an exact time-dependent displacement of the same hierarchy, converged physical-root observables must agree. Its purpose is to reduce the auxiliary basis required when the eliminated environment becomes highly occupied. In large systems a fixed auxiliary cutoff can still cause rare trajectory instabilities, for which adaptive system and auxiliary bases should be used as described in Ref. [9].

VI. BATH OBSERVABLES AFTER ELIMINATING THE MODES

HOPS removes the photon Hilbert spaces from the propagated state, but it does not remove access to physical bath observables. Under the nonlinear measure, the shifted Husimi coordinate of normal mode r is, in our phase convention,

$$q_r(t)=-i\left[z_r(t)+s_r^*(t)\right].\tag{32}$$

Transforming $\boldsymbol{q}$ back to local photon coordinates gives

$$\langle a_i^\dagger a_j\rangle=\mathbb{E}[q_i^*q_j]-\delta_{ij},\qquad \langle a_ia_j\rangle=\mathbb{E}[q_iq_j].\tag{33}$$

Known reference-noise moments can be subtracted trajectory by trajectory as zero-mean control variates. This greatly reduces the variance without changing the estimator. The same conditional construction gives

$$\begin{aligned}\langle n_{q_i}n_{a_j}\rangle&=\mathbb{E}\left[\langle n_{q_i}\rangle_\phi(|q_j|^2-1)\right],\\ \langle S_i^\dagger a_j\rangle&=\mathbb{E}\left[\langle S_i^\dagger\rangle_\phi q_j\right].\end{aligned}\tag{34}$$

The subtraction in Eq. (33) is the Husimi vacuum offset. An auxiliary-state norm is not a photon occupation.

VII. TWO QUBITS AND TWO DAMPED PHOTON MODES

The full benchmark retains the two qubits and two local photon modes,

$$H_S = \omega_q \sum_{i=1}^{2} \sigma_i^+ \sigma_i^-,$$
$$H_B = \omega_c \sum_{i=1}^{2} a_i^\dagger a_i + J_b(a_1^\dagger a_2 + a_2^\dagger a_1), \tag{35}$$

and evolves

$$\dot{\rho} = -i[H_S + H_B + H_I, \rho] + \kappa \sum_{i=1}^{2} \mathcal{D}[a_i]\rho, \qquad \mathcal{D}[a]\rho = a\rho a^\dagger - \frac{1}{2}\{a^\dagger a, \rho\}. \tag{36}$$

There is no direct qubit-qubit term. We compare

$$H_I^{\mathrm{JC}} = g \sum_i (\sigma_i^- a_i^\dagger + \sigma_i^+ a_i), \tag{37}$$

$$H_I^{\mathrm{Rabi}} = g \sum_i (\sigma_i^- + \sigma_i^+)(a_i + a_i^\dagger). \tag{38}$$

The Jaynes-Cummings interaction conserves total excitation before loss. The Rabi interaction additionally contains $\sigma_i^- a_i$ and $\sigma_i^+ a_i^\dagger$, so bare excitation number is not conserved, multi-photon sectors are occupied and anomalous correlations may develop.

Both calculations start in $|e, g, 0, 0\rangle$ and use

$$\omega_q = \omega_c = 1, \qquad g = 0.30, \qquad J_b = 0.40, \qquad \kappa = 0.10. \tag{39}$$

The nonlinear HOPS timestep is 0.02, the total hierarchy depth is seven, and 4096 antithetic trajectories are divided into eight independent batches. The plotted error bars are 95 percent batch intervals. The reference uses two Fock levels per mode for JC, which is exact in its one-excitation sector, and eight levels per mode for Rabi. Increasing the Rabi cutoff from seven to eight changes the reduced state by at most 1.25×10^{-4}, while eight to nine changes it by 2.49×10^{-5}. On identical noise paths, increasing the HOPS depth from seven to nine changes the reduced state by 1.08×10^{-4} over $0 \le t \le 12$ and by 4.58×10^{-4} over $0 \le t \le 24$.

VIII. RESULTS

A. Excitation flow and its return

Figure 2 shows the complete flow over $0 \le t \le 24$. In the JC model the packet moves sequentially from qubit 1 to mode 1, mode 2 and qubit 2. The receiver reaches population 0.7233 at $t = 8.76$. The photon flow subsequently reverses and qubit 1 returns to population 0.4999 at $t = 17.72$. At that time nonlinear HOPS gives 0.5022 ± 0.0070 at 95 percent confidence.

The Rabi flow retains the same broad sequence but is no longer a single conserved packet. A small early receiver feature is produced by counter-rotating dressing. The main receiver peak is 0.6383 at $t = 8.88$, and the returning qubit-1 population reaches 0.3667 at $t = 19.68$; nonlinear HOPS gives 0.3696 ± 0.0060 at the same time. The emitted-photon curve is

$$N_{\mathrm{emit}}(t) = \kappa \int_0^t \mathrm{d}s\, [\langle n_{a_1}(s)\rangle + \langle n_{a_2}(s)\rangle]. \tag{40}$$

It is a loss probability only in the one-excitation JC problem. For Rabi it is a cumulative count because counter-rotating terms change the bare internal excitation number.

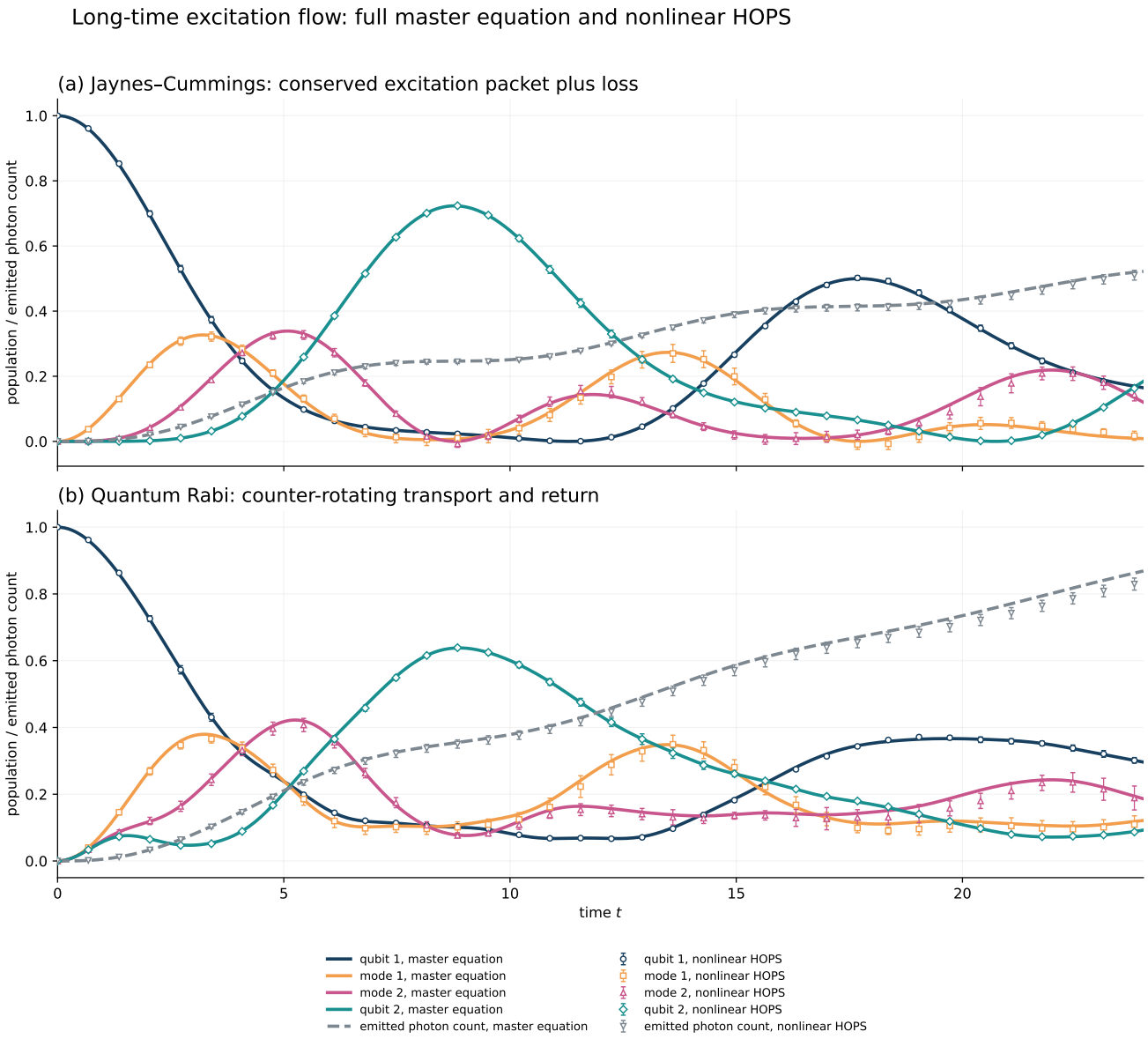


FIG. 2. Long-time excitation flow. Solid and dashed curves are the full master equation; markers and 95 percent intervals are normalized nonlinear HOPS. The calculation without explicit photon modes resolves both the outgoing transfer and the later population return.

The present JC dynamics is slower than the first exploratory calculation, which used $g = 0.45$, $J_b = 0.60$ and $\kappa = 0.15$. All three rates were reduced by the same factor 2/3 for the controlled JC-Rabi comparison, so the time axis stretches by 3/2. The old receiver maximum at $t = 5.85$ therefore predicts $t = 8.775$, in agreement with the present $t = 8.76$.

B. Rotating-wave and counter-rotating physics

The distinction between the interactions is summarized in Fig. 3 and Table 1. JC keeps the total internal excitation below one and has no anomalous photon-pair correlation. Rabi dynamics raises the total internal excitation to 1.1839 and produces $|\langle a_1 a_2 \rangle| = 0.1194$. It also lowers the largest qubit-qubit concurrence from 0.3194 to 0.2065. Nonlinear HOPS tracks each of these changes rather than relying on the special single-excitation structure of the JC model.

TABLE 1. Selected full master-equation observables. The HOPS overlays in the figures reproduce these values within their batch uncertainty.

observable	Jaynes-Cummings	quantum Rabi
maximum receiver population	0.7233	0.6383
time of receiver maximum	8.76	8.88
maximum total internal excitation	1.0000	1.1839
maximum qubit concurrence	0.3194	0.2065
$\max\lvert\langle a_1 a_2\rangle\rvert$	0	0.1194
maximum reduced-state HOPS distance, $t \leq 12$	0.00404	0.00814

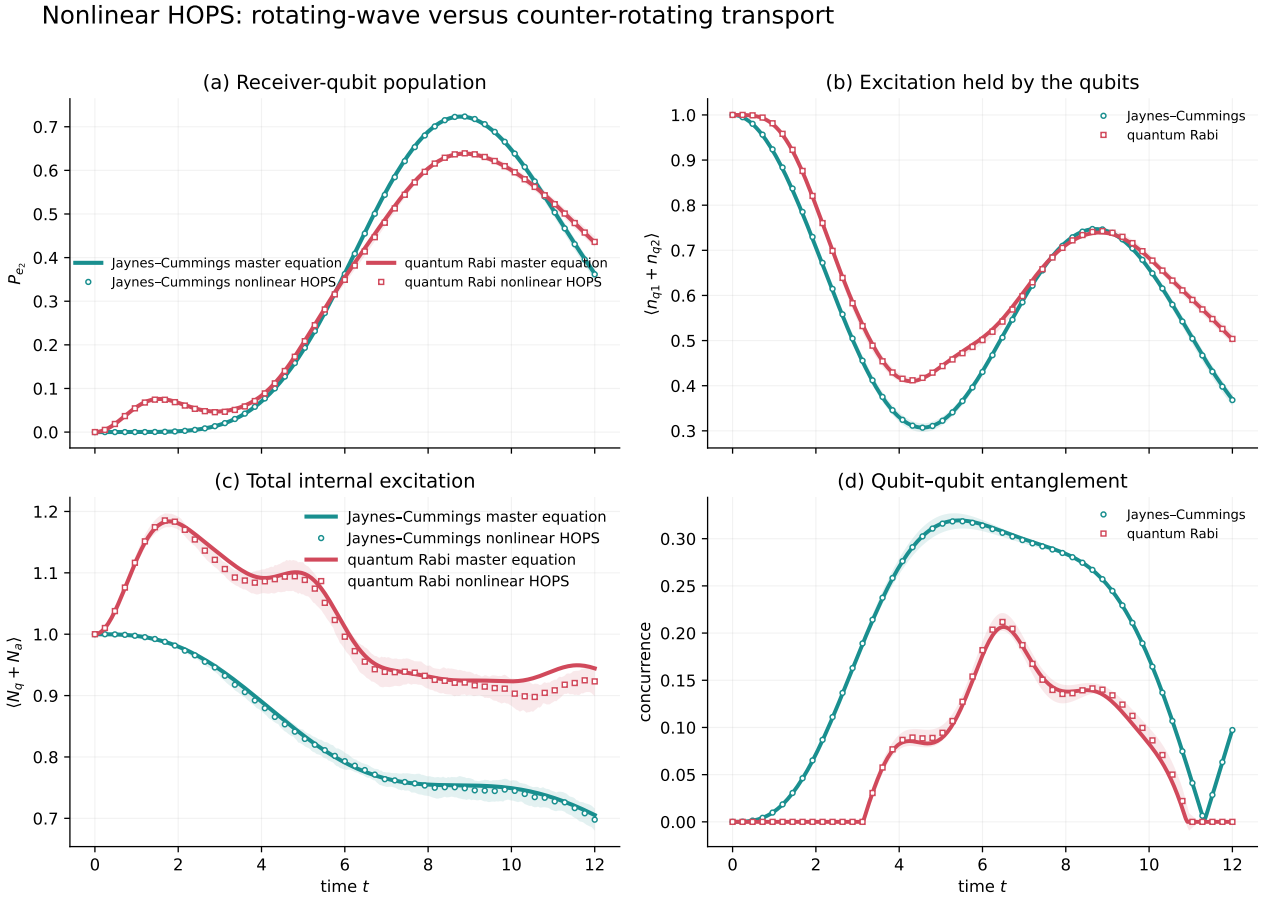


FIG. 3. System observables for the two interactions. Counter-rotating terms alter the receiver transfer, total excitation and qubit entanglement. The master-equation lines and nonlinear-HOPS estimates agree at the stochastic sampling scale.

C. Photon occupations and inter-mode correlations

Figure 4 uses Eq. (33) to reconstruct observables of modes that are absent from the HOPS state. The two occupations reveal the sequential propagation, while the signed current

$$I_{1\to 2} = -2J_b \,\mathrm{Im}\langle a_1^\dagger a_2\rangle \tag{41}$$

changes sign during the return. The anomalous correlation remains zero in JC and becomes finite in Rabi, making it a direct diagnostic of the counter-rotating processes.

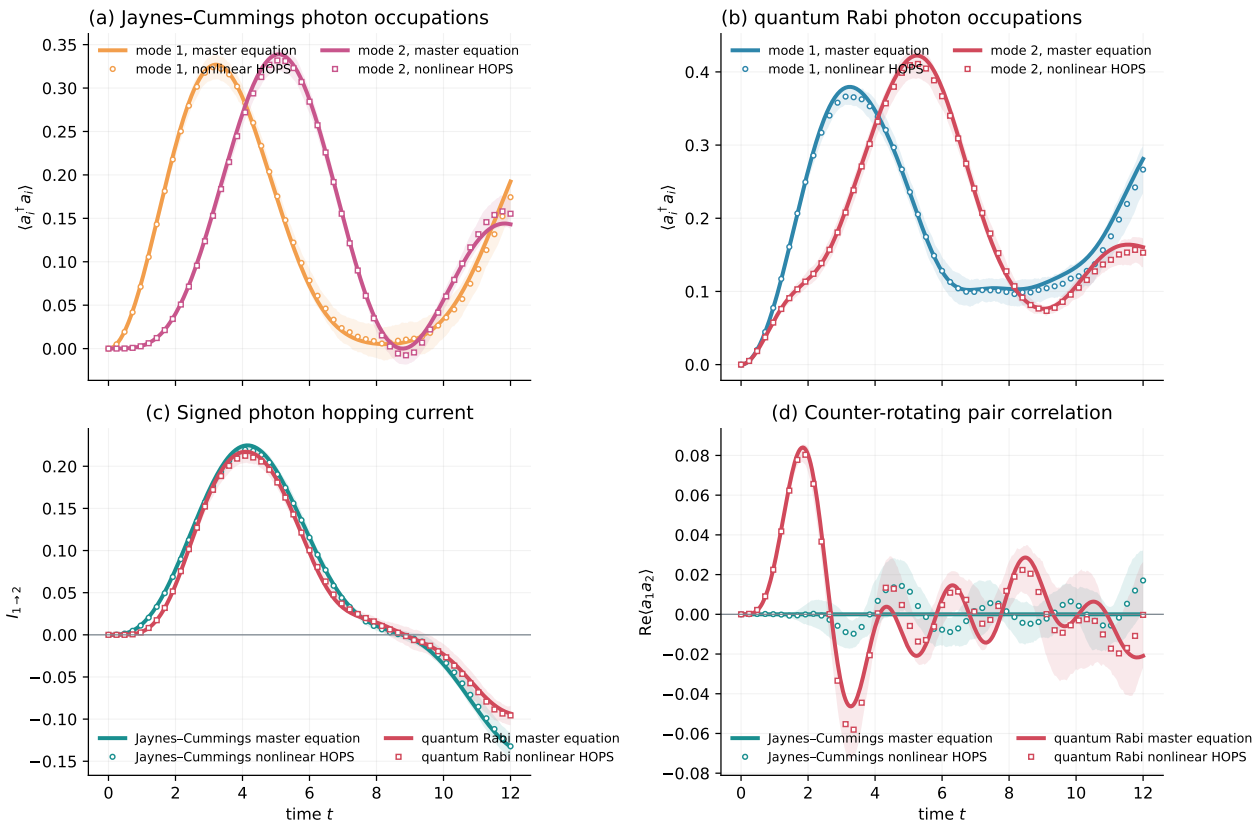


FIG. 4. Physical photon observables reconstructed from nonlinear HOPS. The photon occupations, hopping current and anomalous pair amplitude agree with the explicit-mode reference.

The complete inter-mode two-point comparison is shown in Fig. 5. This includes the real and imaginary parts of $G_{12}^{(1)} = \langle a_1^\dagger a_2 \rangle$, the connected number correlation

$$\langle n_1 n_2 \rangle_c = \langle n_1 n_2 \rangle - \langle n_1 \rangle \langle n_2 \rangle, \tag{42}$$

and $\mathrm{Re}\langle a_1 a_2 \rangle$. Number-number estimators have larger variance than first-order coherences, but no systematic disagreement with the reference is resolved.

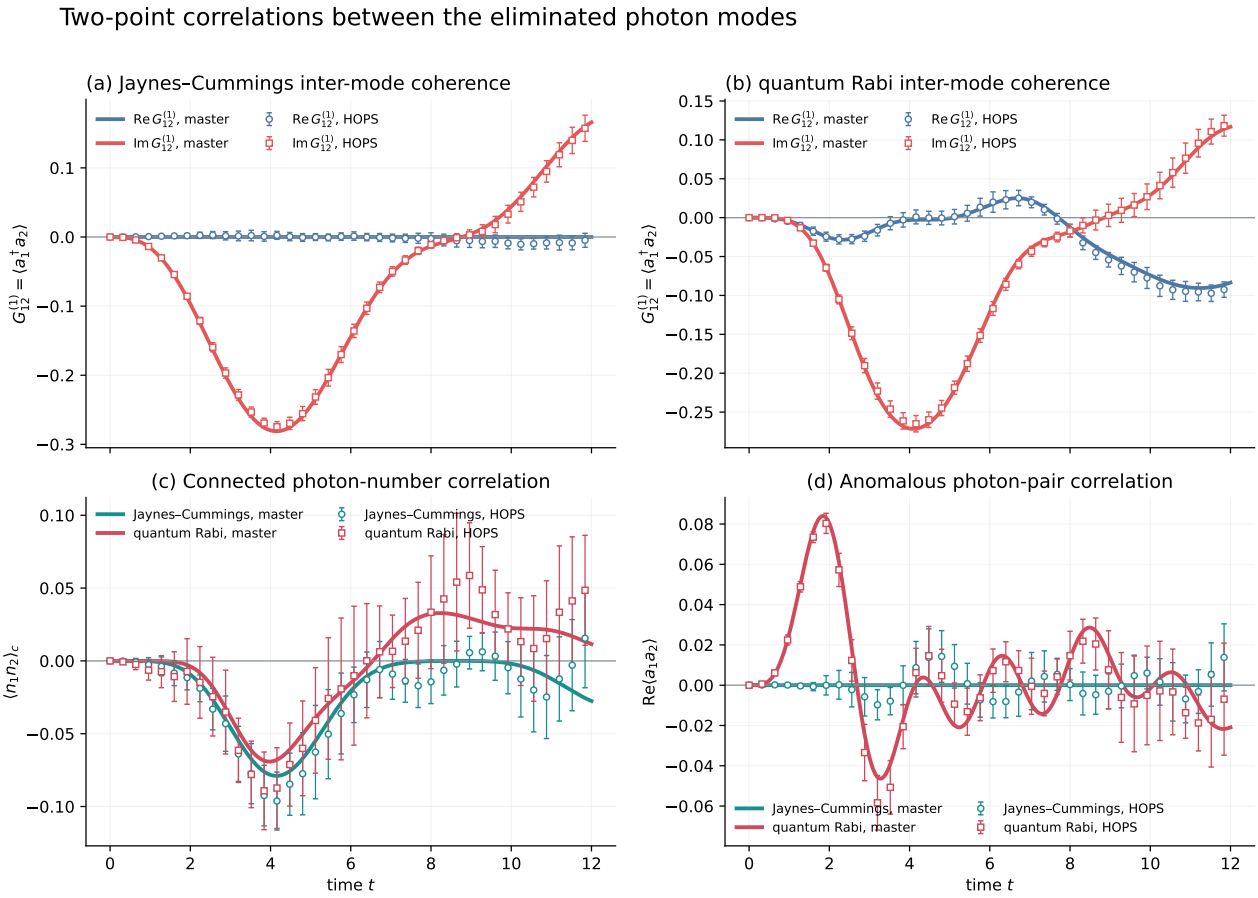


FIG. 5. Two-point correlations between the eliminated photon modes. HOPS recovers normal coherence, connected occupation correlations and the counter-rotating anomalous pair amplitude.

D. Every qubit-mode pair

Finally, Fig. 6 resolves all four combinations of a qubit site and a photon mode. The connected number correlations distinguish local and nonlocal correlations, while $2\,\mathrm{Re}\langle S_i^\dagger a_j \rangle$ follows the coherent exchange channels. These quantities are more demanding than a reduced system population because they mix conditional system averages with noisy Husimi moments. Their agreement is therefore a useful validation that the nonlinear measure and the transformed local photon coordinates have been implemented consistently.

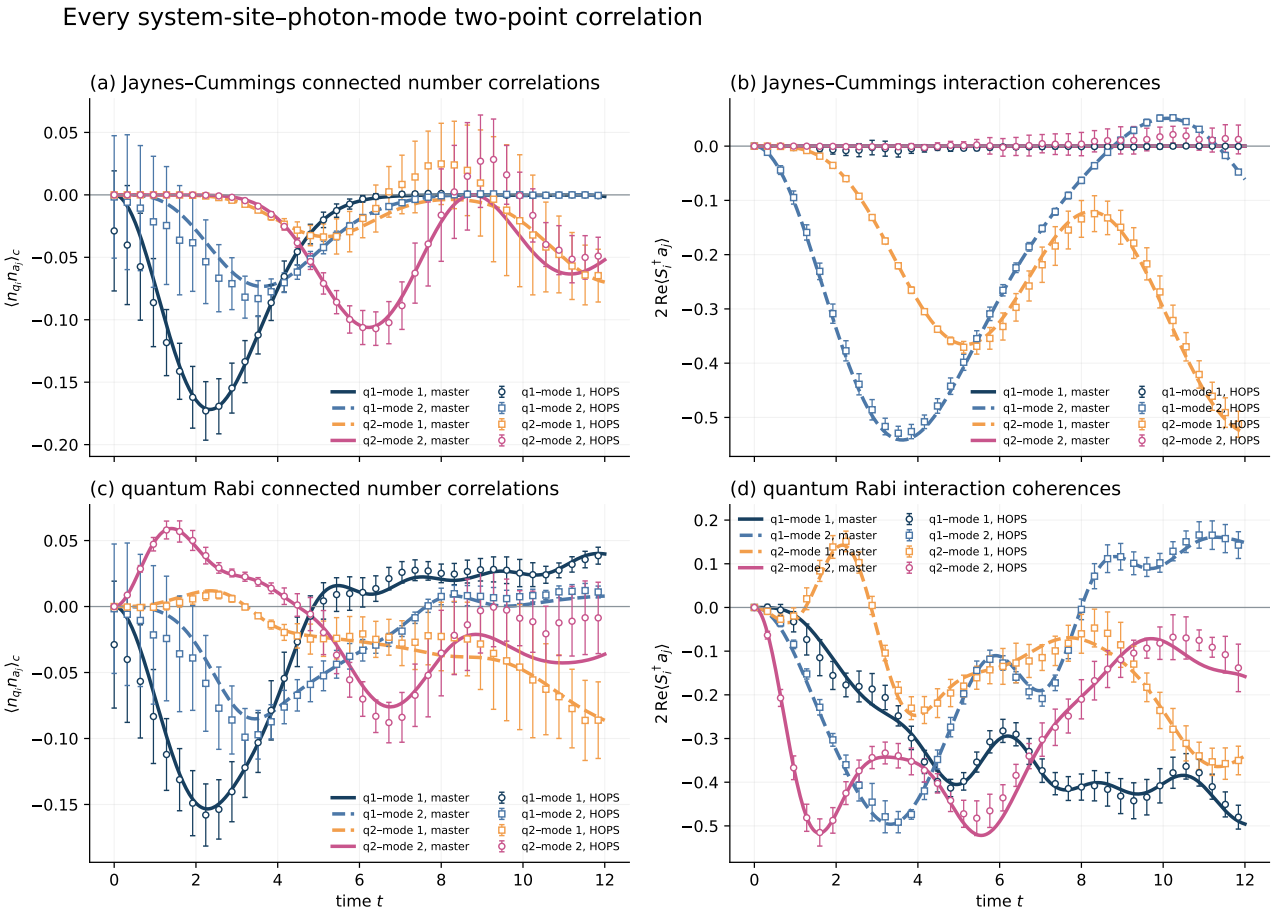


FIG. 6. All qubit-site–photon-mode two-point correlations. The left column shows connected number correlations and the right column shows interaction coherences, for JC and Rabi coupling.

IX. OUTLOOK AND LIMITATIONS

The two-mode problem establishes more than population transfer. A normalized correlated HOPS calculation reproduces reduced states, emitted flux, physical bath occupations and mixed system-bath correlations for both an excitation-preserving model and a model with unrestricted counter-rotating processes. The Rabi calculation is especially useful because it removes the single-excitation simplification: the same hierarchy must describe multi-photon virtual and real processes.

For many-body applications, Eq. (13) is the natural continuation of our MPS-HOPS work [6]. A local or block-local W produces local hopping between hierarchy coordinates, while a normal-mode factorization reduces the number of hierarchy directions when only a few collective bath modes are important. The optimal representation will depend on the spatial range, memory time and tensor-network bond dimension. Useful targets include photon-mediated correlation spreading in cavity arrays, multimode cavity self-organization, waveguide-mediated transport, common-mode noise in circuit QED, phonon-assisted dynamics in molecular or solid-state systems, and correlated reservoirs for quantum simulation and error correction.

Several qualifications remain. The derivation assumes an initially factorized Gaussian bath and linear coupling. A continuum spectral density must be represented by a positive and converged matrix kernel. Hierarchy depth, timestep, stochastic sampling and any MPS compression are independent convergence axes. At finite temperature one may introduce thermofield channels or a thermal coherent-state construction. Non-Gaussian baths and initial system-bath correlations require additional structure.

For the Rabi benchmark, the local dissipator $\kappa\mathcal{D}[a_i]$ is a simple and clearly specified numerical reference. At genuinely ultrastrong coupling a microscopic loss model should normally be constructed in the dressed eigenbasis; bare local damping can convert virtual dressing into spurious steady emission. This does not affect the HOPS-versus-master-equation validation, because both solve the same stated model, but it matters when the cumulative photon count is interpreted as an experimental prediction.

The next methodological step is to combine a positive matrix-kernel fit with the local correlated nuHOPS generator

in Eq. (28) and an MPS system Hamiltonian. This retains the auxiliary hopping of Eq. (14) while following the trajectory-dependent memory displacement. It would extend the 2022 HOPS-MPS method from independent damped modes to environments that actively transport quantum information and correlations between distant parts of an interacting system, including regimes where those environments become highly excited.

X. GENERATING CORRELATED COLOURED NOISE

The stochastic hierarchy requires a proper complex Gaussian vector process $\boldsymbol{z}(t)$ with

$$\mathbb{E}[\boldsymbol{z}(t)\boldsymbol{z}^\dagger(s)] = \alpha(t-s), \qquad \mathbb{E}[\boldsymbol{z}(t)\boldsymbol{z}^T(s)] = 0. \tag{43}$$

The second equation is the vanishing pseudo-covariance. Generating the first covariance is a matrix-factorization problem: spatial cross correlations cannot in general be added as independent pairwise noises.

A. Spectral factorization

Extend the stationary kernel to negative time by $\alpha(-\tau) = \alpha(\tau)^\dagger$ and define the two-sided spectral matrix

$$S(\omega) = \int_{-\infty}^{\infty} \mathrm{d}\tau\, \alpha(\tau) e^{i\omega\tau}, \qquad \alpha(\tau) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \mathrm{d}\omega\, S(\omega) e^{-i\omega\tau}. \tag{44}$$

For a physical stationary Gaussian process, $S(\omega)$ is Hermitian and positive semidefinite at every frequency. In the scalar case the usual filter is simply $H(\omega) = \sqrt{S(\omega)}$. For several correlated channels, replace that scalar square root by a matrix factorization

$$S(\omega) = H(\omega)H^\dagger(\omega). \tag{45}$$

A Cholesky factor may be used when the matrix is positive definite; an eigendecomposition is more robust for a rank-deficient spectrum. The factor is not unique because $H(\omega)U(\omega)$ gives the same spectrum for any unitary $U(\omega)$.

Let $\boldsymbol{\xi}(t)$ contain independent proper complex white noises,

$$\mathbb{E}[\boldsymbol{\xi}(t)\boldsymbol{\xi}^\dagger(s)] = I\delta(t-s), \qquad \mathbb{E}[\boldsymbol{\xi}(t)\boldsymbol{\xi}^T(s)] = 0, \tag{46}$$

and define the matrix response function

$$\mathcal{R}(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \mathrm{d}\omega\, H(\omega) e^{-i\omega t}, \qquad \boldsymbol{z}(t) = \int_{-\infty}^{\infty} \mathrm{d}s\, \mathcal{R}(s)\boldsymbol{\xi}(t-s). \tag{47}$$

Using Eq. (46) and the convolution theorem gives

$$\begin{aligned}\mathbb{E}[\boldsymbol{z}(t)\boldsymbol{z}^\dagger(s)] &= \int \mathrm{d}u\, \mathcal{R}(u)\mathcal{R}^\dagger(u-t+s) \\ &= \frac{1}{2\pi}\int \mathrm{d}\omega\, H(\omega)H^\dagger(\omega)e^{-i\omega(t-s)} = \alpha(t-s).\end{aligned} \tag{48}$$

Thus one needs one independent white-noise vector, whose dimension is the rank of $S(\omega)$, rather than a separate noise for every pair of sites.

B. Discrete frequency grid and FFT

Choose N frequencies $\omega_k = \omega_{\min} + k\Delta\omega$ and times $t_j = j\Delta t$ with $N\Delta\omega\Delta t = 2\pi$. For independent vectors $\boldsymbol{\xi}_k \sim \mathcal{CN}(\boldsymbol{0}, I)$, a midpoint discretization is

$$\boxed{\boldsymbol{z}(t_j) = \sqrt{\frac{\Delta\omega}{2\pi}}\sum_{k=0}^{N-1} H(\omega_k)\boldsymbol{\xi}_k e^{-i\omega_k t_j}.} \tag{49}$$

Its discrete covariance is

$$\mathbb{E}[\boldsymbol{z}(t_j)\boldsymbol{z}^\dagger(t_\ell)] = \frac{\Delta\omega}{2\pi}\sum_k S(\omega_k)e^{-i\omega_k(t_j-t_\ell)}, \tag{50}$$

which converges to Eq. (44). Equation (49) is evaluated with one FFT for each output channel, with the normalization adjusted to the FFT convention.

In practice, first symmetrize each numerical spectral matrix, $S_k \leftarrow (S_k + S_k^\dagger)/2$, then diagonalize it. Small negative eigenvalues at the level of floating-point error may be clipped to zero; resolved negative eigenvalues indicate an invalid kernel, insufficient frequency resolution or an inconsistent Fourier convention and must not be hidden. The frequency window must be wide enough that the omitted spectral weight is negligible, while the total simulated interval must be short enough that FFT periodicity does not wrap correlations back into the data.

C. Exact recursion for a matrix-exponential kernel

For the kernels used in this note there is a cheaper time-domain construction. Let

$$\alpha(\tau) = Re^{-W\tau}R^\dagger, \qquad W = \frac{K}{2} + ih, \qquad K = W + W^\dagger \succeq 0, \qquad \tau \geq 0. \tag{51}$$

Generate a stationary multivariate complex Ornstein–Uhlenbeck process

$$\mathrm{d}\boldsymbol{x} = -W\boldsymbol{x}\,\mathrm{d}t + B\,\mathrm{d}\boldsymbol{\beta}, \qquad BB^\dagger = K, \tag{52}$$

where $\mathbb{E}[\mathrm{d}\boldsymbol{\beta}\,\mathrm{d}\boldsymbol{\beta}^\dagger] = I\,\mathrm{d}t$ and $\mathbb{E}[\mathrm{d}\boldsymbol{\beta}\,\mathrm{d}\boldsymbol{\beta}^T] = 0$. With $\boldsymbol{x}(0) \sim \mathcal{CN}(\mathbf{0}, I)$,

$$\mathbb{E}[\boldsymbol{x}(t)\boldsymbol{x}^\dagger(s)] = e^{-W(t-s)} \qquad (t \geq s). \tag{53}$$

The site noises are therefore

$$\boldsymbol{z}(t) = R\boldsymbol{x}(t), \tag{54}$$

which immediately gives Eq. (51).

No time-discretization approximation is needed for this linear process. For a step Δt, set

$$F = e^{-W\Delta t}, \qquad Q = I - FF^\dagger, \qquad Q = CC^\dagger, \tag{55}$$

and update

$$\boxed{\boldsymbol{x}_{n+1} = F\boldsymbol{x}_n + C\boldsymbol{\eta}_n, \qquad \boldsymbol{\eta}_n \sim \mathcal{CN}(\mathbf{0}, I).} \tag{56}$$

The equality $Q \succeq 0$ follows from $K \succeq 0$. This recursion preserves the unit stationary covariance exactly. For the coupled-cavity example, $R = gI$, $W = \kappa I/2 + ih_B$, and the same W is used in the noise, nonlinear history shift and hierarchy drift.

The process in Eqs. (49) or (56) is the reference Gaussian noise. The trajectory-dependent Girsanov shift is added while propagating nonlinear HOPS or nuHOPS; the shifted process should not be generated as an independent stationary Gaussian field.

D. Why an elementwise pair construction fails

Taking $\sqrt{S_{ij}(\omega)}$ entry by entry does not produce a matrix H satisfying Eq. (45). If a shared noise is assigned to a pair (i, j), its contribution to a diagonal spectrum is the modulus squared of its filter. It is therefore not consistent to define a residual diagonal spectrum as $S_{ii} - \sum_{j\neq i} S_{ij}$: that quantity need not be real or positive, and for many spatial covariance matrices it becomes negative. The pairwise construction also cannot enforce the global positive-semidefinite constraint or, in its real symmetric form, retain the phases of complex cross spectra.

Such a decomposition works only for a restricted class of real, nonnegative, diagonally dominant spectra with specially chosen pair filters. Equations (45) and (56) are the general constructions. A numerical implementation should verify the empirical covariance at several positive and negative lags, the Hermitian relation $\alpha(-\tau) = \alpha(\tau)^\dagger$, and the vanishing pseudo-covariance before the noise is used in a HOPS calculation.

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